On tree-based methods for (partial) differential equations

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(joint work with Guillaume Penent and Jiang Yu Nguwi)

Stochastic branching mechanisms have been used to represent the solutions of partial differential equations in [15], [7], [10], [8], and recently extended in [6] to the treatment of polynomial nonlinearities in first order gradient terms. This talk reviews an extension of such tree-based methods to functional nonlinearities with gradients of arbitrary orders.

Consider the ODE

$$u'(t) = f(u(t)), \qquad u(0) = u_0 \in \mathbb{R}^d, \quad t \in \mathbb{R}_+, \tag{1}$$

whose solution can be expanded as

$$u(t) = u_0 + tf(u_0) + \frac{t^2}{2} f'f(u_0) + \frac{t^3}{6} f'f'f(u_0) + \frac{t^3}{6} f''[f,f](u_0) + \cdots$$

which rewrites as the sum

$$u(t) = u_0 + \sum_{\mathcal{T}} \frac{t^{r(\mathcal{T})}}{\sigma(r(\mathcal{T}))\gamma(r(\mathcal{T}))} F(\mathcal{T})$$

over the family of Butcher trees $\mathcal{T}$, see [1], [2], Chapters 4-6 of [4], and [9], based on early work of [3]. In order to solve (1), we may also write

$$u(s) = u_0 + \int_0^s u'(r)dr = u_0 + \int_0^s f(u(r))dr,$$

and more generally we can expand the derivative $f^{(l)}(u(r))$ as

$$f^{(l)}(u(r)) = f^{(l)}(u_0) + \int_0^r f(u(r)) f^{(l+1)}(u(v))dv, \qquad l \geq 1.$$

We note that the above family of equations can be rewritten as

$$c(u)(t) = c(u)(0) + \sum_{Z \in \mathcal{M}(c)} \int_0^t \prod_{z \in Z} z(u)(s)ds \tag{2}$$

where c runs through a set $\mathcal{C} := \left\{ \mathrm{Id},\ f^{(l)},\ l \geq 0 \right\}$, of functions called *codes* and $\mathcal{M}(c)$ is defined by letting $\mathcal{M}(\mathrm{Id}) := \{f\}$ and $\mathcal{M}(g) := \left\{(f, g')\right\}$ for g a smooth function on $\mathbb{R}_+ \times \mathbb{R}$, see [12].

Next, consider a nonlinear PDE of the form

$$\begin{cases} \partial_t u(t,x) + \dfrac{1}{2}\Delta u(t,x) + f(u(t,x)) = 0 \\ u(T,x) = \phi(x), \qquad (t,x) \in [0,T] \times \mathbb{R}. \end{cases} \tag{3}$$

Letting $v(t,x) := g(u(t,x))$, we now have

$$\partial_t v(t,x) + \frac{1}{2}\Delta v(t,x) = g'(u(t,x))\left(\partial_t u(t,x) + \frac{1}{2}\Delta u(t,x)\right) + \frac{1}{2}(\partial_x u(t,x))^2 g''(u(t,x))$$
$$= -f(u(t,x))g'(u(t,x)) + \frac{1}{2}(\partial_x u(t,x))^2 g''(u(t,x)),$$

which shows that the functions u, $\partial_x u$, $af^{(k)} \circ u$ satisfy the integral equations

$$\begin{cases} u(t,x) = \displaystyle\int_{-\infty}^{\infty} \varphi(T-t, y-x)\phi(y)dy + \int_t^T \int_{-\infty}^{\infty} \varphi(s-t, y-x) f(u(s,y))dyds \\ af^{(k)}(u(t,x)) = \displaystyle\int_{-\infty}^{\infty} \varphi(T-t, y-x) af^{(k)}(\phi(y))dy \\ + \displaystyle\int_t^T \int_{-\infty}^{\infty} \varphi(s-t, y-x) \\ \quad \times \left(af(u(s,y))f^{(k+1)}(u(s,y)) - \frac{a}{2}(\partial_x u(s,y))^2 f^{(k+2)}(u(s,y))\right) dyds \\ \partial_x u(t,x) = \displaystyle\int_{-\infty}^{\infty} \varphi(T-t, y-x)\partial_x\phi(y)dy \\ \quad + \displaystyle\int_t^T \int_{-\infty}^{\infty} \varphi(s-t, y-x) f'(u(s,y))\partial_x u(x,y)dyds, \end{cases}$$

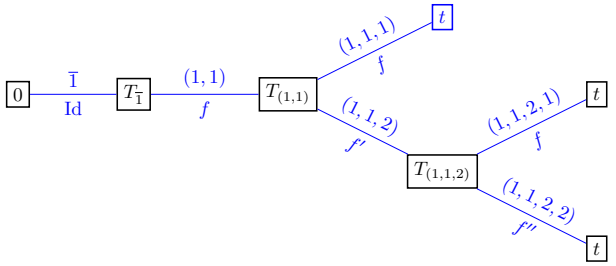
$a \neq 0$, $k \in \mathbb{N}$. We note that the above set of equations admits a formulation identical to (2) provided that we use the codes

$$\mathcal{C} := \left\{\mathrm{Id},\ \partial_x,\ af^{(k)},\ a \neq 0,\ k \in \mathbb{N}\right\}$$

and the mechanism defined as

$$\mathcal{M}(\mathrm{Id}) := \{f^*\}, \quad \mathcal{M}(g^*) := \left\{(f^*, (g')^*), \left(\partial_x, \partial_x, -\frac{1}{2}(g'')^*\right)\right\},$$

and $\mathcal{M}(\partial_x) := \{((f')^*, \partial_x)\}$.



We consider a random coding tree $\mathcal{T}_{t,x,c}$ illustrated by the above sample, started at (t,x) with a code $c \in \mathcal{C}$ and partitioned as $\mathcal{K}^\partial \cup \mathcal{K}^\circ$, where $\mathcal{K}^\circ$ denotes the set of leaves. In the next result, we use the random functional

$$\mathcal{H}(\mathcal{T}_{t,x,c}) := \prod_{\overline{k} \in \mathcal{K}^\circ} \frac{1}{q_{c_{\overline{k}}}\rho(\tau_{\overline{k}})} \prod_{\overline{k} \in \mathcal{K}^\partial} \frac{c_{\overline{k}}(u)(T, X^{\overline{k}}_{T_{\overline{k}}})}{\overline{F}(T - T_{\overline{k}-})}.$$

of the random coding tree $\mathcal{T}_{t,x,c}$, in which branching at a node $\overline{k}$ occurs at the random time $T_{\overline{k}}$, the interjump time $\tau_{\overline{k}} = T_{\overline{k}} - T_{\overline{k}-}$ has tail CDF $\overline{F}$ and PDF ρ, and $\left(X_t^{\overline{k}}\right)_{t \geq T_{\overline{k}-}}$ is an independent Brownian motion started at time $T_{\overline{k}-}$.

Theorem 1. *([13]) Assume that the integral solution of the system* (2) *is unique and that there exists a constant* $K > 0$ *such that:*

$$|f^{(k)} \circ \phi|_\infty \leq K, \quad k \geq 0, \quad |\phi|_\infty \leq K, \quad |\phi'|_\infty \leq K.$$

Then, there exists $T > 0$ *such that the solution of* (3) *admits the probabilistic representation*

$$u(t,x) = \mathbb{E}\left[\mathcal{H}(\mathcal{T}_{t,x,\mathrm{Id}})\right], \qquad (t,x) \in [0,T] \times \mathbb{R}.$$

The above method also extends to fully nonlinear PDEs of the form

$$\begin{cases} \partial_t u(t,x) + \dfrac{1}{2}\Delta u(t,x) + f\big(u(t,x), \nabla u(t,x), \ldots, \nabla^n u(t,x)\big) = 0, \\ u(T,x) = \phi(x), \qquad (t,x) = (t,x_1,\ldots,x_d) \in [0,T] \times \mathbb{R}^d, \end{cases}$$

$d \geq 1$, see [13], [11]. As an example, we consider a cosine nonlinearity with a gradient of order four, for which our method appears more accurate than the deep Galerkin method [14]. Related comparisons can be found in [11] with respect to the deep BSDE method [5].

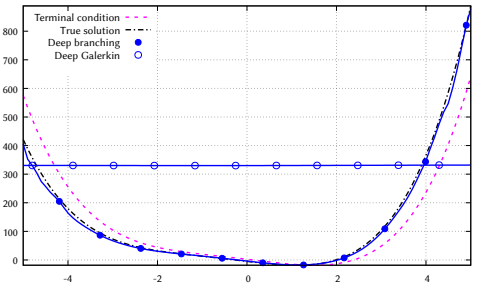


FIGURE 1. Comparison graphs in dimension $d = 5$.

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